

Physical Chemistry

Paper - 2023

Subjective Type

Section - I

Question - 1

-(i)-

Determination of pre-exponential factor:
According

to collision theory:

$$K_2 = DT^{1/2} e^{-\frac{E^*}{RT}}$$

By taking natural log on B.S

$$\ln K_2 = \ln D + \frac{1}{2} \ln T - \frac{E^*}{RT}$$

$$\ln K_2 = \ln D - \frac{1}{2} \ln \left(\frac{1}{T} \right) - \frac{E^*}{R} \left(\frac{1}{T} \right) \text{---(i)}$$

By differentiating eq. no: (i) w.r.t. $\left(\frac{1}{T} \right)$

$$\frac{d \ln K_2}{d \left(\frac{1}{T} \right)} = \frac{d}{d \left(\frac{1}{T} \right)} \ln D - \frac{1}{2} \frac{d}{d \left(\frac{1}{T} \right)} \ln \frac{1}{T} - \frac{E^*}{R} \frac{d}{d \left(\frac{1}{T} \right)} \left(\frac{1}{T} \right)$$

$$\frac{d \ln k_2}{d\left(\frac{1}{T}\right)} = 0 - \frac{1}{2} \left(\frac{1}{T}\right) - \frac{E^*}{R} \quad (1)$$

$$\frac{d \ln k_2}{d\left(\frac{1}{T}\right)} = -\frac{1}{2} T - \frac{E^*}{R} \quad (2)$$

Multiply eq no. (ii) by $-R$.

$$-R \frac{d \ln k_2}{d\left(\frac{1}{T}\right)} = \frac{RT}{2} + E^*$$

$$\frac{RT}{2} + E^* = -R \frac{d \ln k_2}{d\left(\frac{1}{T}\right)} \quad (3)$$

By comparing eq no. (iii) with Arrhenius equation;

$$E_a = E^* + \frac{RT}{2}$$

$$E^* = E_a - \frac{RT}{2}$$

By putting value of E^* in equation;

$$k_2 = P d^2 \left[\frac{8\pi RT (M_A + M_B)}{M_A M_B} \right]^{1/2} e^{-\frac{E^*}{RT}}$$

$$k_2 = P d^2 \left[\frac{8\pi RT (M_A + M_B)}{M_A M_B} \right]^{1/2} e^{1/2} e^{-\frac{E_a}{RT}}$$

By comparing with $k = A e^{-\frac{E_a}{RT}}$ we get

$$A = P d^2 \left[\frac{8\pi RT (M_A + M_B)}{M_A M_B} \right]^{1/2} e^{1/2}$$

—(ii)—

Effect of temperature on rate constant:-

The rate constant of a reaction is a function of temperature;

Any factor that affects rate constant also affects the rate of a reaction.

With increase of temperature the rate of reaction and rate constant also increases. As a generalization, the rate constant becomes almost **double** for every **10 degree** rise in temperature. This is also called as temperature coefficient.

—(iii)—

Ionic Reactions:-

Ionic reaction is generally the reaction of ions i.e. anions and cations, to form a compound in a medium in which they are dissolved. When ions of

water-soluble salts interact with each other in aqueous medium. It results in the formation of water-insoluble salts.

e.g.

When HCl mix with aq. NaOH, it results in the formation of salt NaCl and H_2O .



-(iv)-

Cage Effect:-

Definition:-

"A phenomenon involving the dissociation of molecules unable to move apart rapidly because of presence of other molecules, with the result that the dissociation products may not recombine, called as cage effect."

⇒ Cage effect describes how the solvent affects the reactants, as they try to come together to

to form the product. Since the solvent molecules predominate, they surround the reactants in a cage-like fashion.

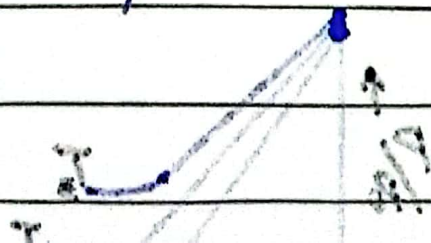
— (xv) —

Microstate:-

A microstate is a specific microscopic configuration of a thermodynamic system that the system may occupy with certain probability in the course of thermal fluctuations.

Description:-

A microstate of a system may be defined by the specification of six coordinates x, y, z, v_x, v_y, v_z of each molecule of the system within the limits of the dimensions of the cell in which its productive point lies.



— (iv) —

Effect of temperature on vertical distribution of gas:-

By Barometric mathematical calculation

Solution is drawn as:

$$P = P_0 e^{-\frac{mgh}{RT}}$$

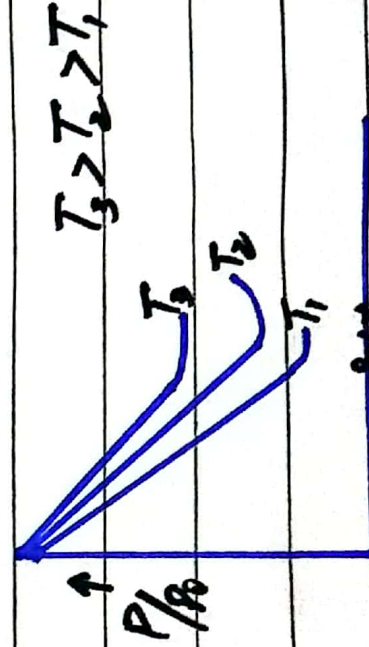
$$\frac{P}{P_0} = e^{-\frac{mgh}{RT}}$$

Description:-

According to above equation the value of $\frac{P}{P_0}$ at given height will be larger at higher temperature than that of lower temperature. Thus, variation in pressure is less gradual at higher temperature.

Graph:-

Graphical explanation can be illustrated as:-



— (xiv) —

Clausius - Inequality:-

Clausius - inequality

tells:

that cyclic integral of $\frac{dQ}{T}$ for a natural cyclic or irreversible cycle $\oint$ is always less than zero.

⇒ It also represent that the cyclic integral of $\frac{dQ}{T}$ of a Reversible cycle with same end point is.

Mathematically:-

$$\oint \frac{dQ}{T} < \oint \frac{dQ}{T}$$

From Clausius theorem:

$$\oint_{rev} \frac{dQ}{T} = 0$$

$$\oint_{irrev} \frac{dQ}{T} < 0$$

—(vic)—

Reaction in Solution phase faster than in gas phase:-

Reactions in solution phase is always faster than in gas phase due to reason.

Reason:-

Solvent cage effect is responsible for this phenomenon. Molecules of different reactants are in cage of molecules of the solvent and collide with each other many time and are converted into products in short interval of time.

But in case of gas phase reactions the probability of collision of the molecules of reactants is less as compared to that in solution.

- (v) -

Ideal gases:-

Definition:-

"Ideal gases are those that obey all the gas laws like Boyle's law, Charles law, Avogadro's law etc under all given temperature and pressure."

Examples:-

H_2 , He , N_2 are ideal at low pressure and higher temperature. In ideal gases there is less or no force of attraction between gas molecules.

- (xii) -

Degree of freedom:-

It is defined

as:

"The total number of coordinates required to describe completely the position of a system."

Example:-

CO Have seven degree of freedom.

* - 3 - degree of freedom for translational motion.

* - 2 - for rotational motion.

* - 2 - for vibrational motion

Thus, total degree of freedom for carbon monoxide is seven.

— (vii) —

Most Probable velocity:-

It is

defined as:

"The velocity possessed by maximum number of molecules of gas at given temperature."

Elucidation:-

The expression for most probable velocity can be derived mathematically by using condition of maxima by setting first derivatives of $\frac{1}{n} \cdot \frac{dnc}{dc}$ with respect to speed equal to zero.

Expression:

$$C_{mp} = 1.414 \sqrt{\frac{RT}{M}}$$

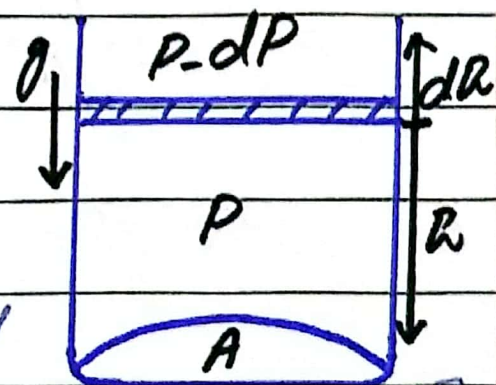
-(xiii)-

Mathematical formulation of
Barometric formula:-

Barometric formula deals with variation of pressure with height as result of influence of gravitational field.

Mathematically:-

Let 'P' be pressure at height 'h' from bottom of the cylinder and $P+dP$ be pressure at height $h+dh$.



$$F_{\text{upward}} = A \times dP$$

$$F_{\text{downward}} = A \times h \times \rho \times g$$

At mechanical equilibrium::

$$F_u = F_D$$

$$\Delta P = \Delta h \rho g$$

$$-dP = dh \rho g$$

(-) sign shows pressure decreases with altitude

$$dP = -\rho g dh$$

As we know that

$$PV = nRT$$

$$PV = \frac{m}{M} RT$$

$$\frac{PM}{RT} = \rho$$

$$dP = -\frac{PM}{RT} g dh$$

$$\frac{dP}{P} = -\frac{Mg}{RT} dh \quad \text{--- (i)}$$

By Integrating eq no: (i)

$$\ln P - \ln P_0 = -\frac{Mg}{RT} (h - 0)$$

$$\ln \frac{P}{P_0} = -\frac{Mgh}{RT}$$

Take Antilog on B.S, we get;

$$\frac{P}{P_0} = e^{\frac{-Mgh}{RT}}$$

$$P = P_0 e^{\frac{-Mgh}{RT}}$$

Two Statements of 2nd Law of thermodynamics:-

i- Clausius Statement:-

Clausius

states that;

"Heat can't spontaneously flow from colder body to hotter body without external work being performed on the system."

ii- Kelvin-Planck Statement:-

It

states that;

"No process is possible in which the sole result is the absorption of heat from reservoir and its complete conversion into work."

Nernst approximation formula :-

Nernst approximation formula is used to estimate the equilibrium potential for an ion across a cell membrane.

Mathematically:

$$E = \frac{RT}{zF} \ln \left[\frac{(\text{ion})_{\text{inside}}}{(\text{ion})_{\text{outside}}} \right]$$

where;

- ⇒ E is equilibrium potential.
- ⇒ R is ideal gas constant
- ⇒ T is absolute temperature
- ⇒ z is valency of ions
- ⇒ F is Faraday's constant.

This formula is based on Nernst equation and provide an estimation when conc. gradient is not extremely large.

— (viii) —

Root Mean Square velocity
from Maxwell's Law:—

The
mean square velocity is defined
as:

$$\bar{C}^2 = C_1^2 + C_2^2 + C_3^2 + \dots + C_n^2$$

$$\bar{C}^2 = \frac{1}{n} \sum_{i=1}^n C_i^2$$

$$\bar{C}^2 = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^{\infty} e^{-\frac{mc^2}{2kT}} c^4 dc$$

$$\bar{C}^2 = 4\pi \left(\frac{1}{\alpha^2 \pi} \right)^{3/2} \int_0^{\infty} e^{-\frac{c^2}{\alpha^2}} c^4 dc$$

$$\bar{C}^2 = 4\pi \left(\frac{1}{\alpha^2 \pi} \right)^{3/2} \cdot \frac{3}{8} \alpha^5 \sqrt{\pi}$$

$$\bar{C}^2 = \frac{4\pi}{\alpha^3 \pi^{3/2}} \cdot \frac{3}{8} \alpha^5 \pi^{1/2}$$

$$\bar{C}^2 = \frac{3}{2} \alpha^2$$

By taking square root on BS

$$\sqrt{\bar{C}^2} = \sqrt{\frac{3}{2} \alpha^2}$$

$$C_{r.m.s} = 1.732 \sqrt{\frac{RT}{M}}$$